

## Subject- DSA

Question 1: Which OOP concept allows the same method name with different implementations?

Option A: Encapsulation

Option B: Inheritance

Option C: Polymorphism

Option D: Abstraction

Correct Answer: Polymorphism

Explanation: Polymorphism allows methods with the same name to perform different actions depending on the object.

Difficulty Level: M

Question 2: Which type of inheritance is not directly supported in Java using classes?

Option A: Single

Option B: Multilevel

Option C: Hierarchical

Option D: Multiple

Correct Answer: Multiple

Explanation: Java does not support multiple inheritance through classes to avoid ambiguity.

Difficulty Level: M

Question 3: What is runtime polymorphism achieved through?

Option A: Method Overloading

Option B: Constructor Overloading

Option C: Method Overriding

Option D: Encapsulation

Correct Answer: Method Overriding

Explanation: Runtime polymorphism occurs when a subclass overrides a parent class method.

Difficulty Level: M

Question 4: Which keyword prevents method overriding?

Option A: static

Option B: final

Option C: const

Option D: private

Correct Answer: final

Explanation: A final method cannot be overridden by subclasses.

Difficulty Level: E

Question 5: Which principle hides implementation details from users?

Option A: Inheritance

Option B: Abstraction

Option C: Polymorphism

Option D: Coupling

Correct Answer: Abstraction

Explanation: Abstraction hides internal implementation and shows only essential features.

Difficulty Level: E

Question 6: Which access modifier allows access only within the same class?

Option A: protected

Option B: public

Option C: private

Option D: default

Correct Answer: private

Explanation: Private members can only be accessed within the same class.

Difficulty Level: E

Question 7: Which feature increases code reusability?

Option A: Encapsulation

Option B: Inheritance

Option C: Constructor

Option D: Destructor

Correct Answer: Inheritance

Explanation: Inheritance allows a class to reuse properties and methods of another classes

Difficulty Level: E

Question 8: Which class is the parent of all classes in Java?

Option A: Main

Option B: Parent

Option C: Object

Option D: Super

Correct Answer: Object

Explanation: Every Java class inherits directly or indirectly from Object.

Difficulty Level: E

Question 9: What is early binding also called?

Option A: Runtime Binding

Option B: Dynamic Binding

Option C: Static Binding

Option D: Late Binding

Correct Answer: Static Binding

Explanation: Early binding occurs during compilation and is known as static binding.

Difficulty Level: M

Question 10: Which method cannot be overridden?

Option A: public

Option B: protected

Option C: static

Option D: abstract

Correct Answer: static

Explanation: Static methods belong to the class and cannot be overridden.

Difficulty Level: M

Question 11: Constructor overloading is an example of:

Option A: Runtime Polymorphism

Option B: Compile-time Polymorphism

Option C: Inheritance

Option D: Abstraction

Correct Answer: Compile-time Polymorphism

Explanation: Constructor overloading is resolved during compilation.

Difficulty Level: M

Question 12: Which OOP principle binds data and methods together?

Option A: Inheritance

Option B: Encapsulation

Option C: Polymorphism

Option D: Overloading

Correct Answer: Encapsulation

Explanation: Encapsulation combines data and methods into a single unit.

Difficulty Level: E

Question 13: Which keyword refers to the current object?

Option A: super

Option B: current

Option C: this

Option D: self

Correct Answer: this

Explanation: The keyword 'this' refers to the current object instance.

Difficulty Level: E

Question 14: Which keyword calls the parent class constructor?

Option A: this

Option B: parent

Option C: base

Option D: super

Correct Answer: super

Explanation: The super keyword invokes the immediate parent constructor.

Difficulty Level: E

Question 15: Which concept allows one interface with many implementations?

Option A: Encapsulation

Option B: Polymorphism

Option C: Constructor

Option D: Coupling

Correct Answer: Polymorphism

Explanation: Polymorphism allows the same interface to have multiple implementations.

Difficulty Level: M

Question 16: What is an abstract class?

Option A: Class with only constructors

Option B: Class without objects

Option C: Class with abstract methods

Option D: Final class

Correct Answer: Class with abstract methods

Explanation: An abstract class can contain abstract and concrete methods.

Difficulty Level: M

Question 17: Which keyword is used to inherit a class in Java?

Option A: inherit

Option B: implements

Option C: extends

Option D: super

Correct Answer: extends

Explanation: The extends keyword creates inheritance between classes.

Difficulty Level: E

Question 18: Which relationship represents HAS-A?

Option A: Inheritance

Option B: Aggregation

Option C: Polymorphism

Option D: Abstraction

Correct Answer: Aggregation

Explanation: Aggregation describes a HAS-A relationship between objects.

Difficulty Level: M

Question 19: Which relationship represents IS-A?

Option A: Aggregation

Option B: Association

Option C: Inheritance

Option D: Composition

Correct Answer: Inheritance

Explanation: Inheritance represents an IS-A relationship.

Difficulty Level: E

Question 20: Which method is automatically called when an object is created?

Option A: finalize()

Option B: delete()

Option C: constructor

Option D: main()

Correct Answer: constructor

Explanation: Constructors are automatically executed when an object is created.

Difficulty Level: E

Question 21: What is function overriding?

Option A: Same method with different parameters

Option B: Redefining parent method in child class

Option C: Multiple constructors

Option D: Hiding variables

Correct Answer: Redefining parent method in child class

Explanation: Overriding allows a subclass to provide a specific implementation of a parent method.

Difficulty Level: M

Question 22: Which cannot be instantiated?

Option A: Object class

Option B: Abstract class

Option C: Child class

Option D: Static class

Correct Answer: Abstract class

Explanation: Abstract classes cannot be instantiated directly.

Difficulty Level: E

Question 23: Which feature supports data security?

Option A: Encapsulation

Option B: Inheritance

Option C: Overloading

Option D: Polymorphism

Correct Answer: Encapsulation

Explanation: Encapsulation protects data through access control.

Difficulty Level: E

Question 24: Which of the following supports dynamic method dispatch?

Option A: Overloading

Option B: Overriding

Option C: Constructors

Option D: Encapsulation

Correct Answer: Overriding

Explanation: Dynamic method dispatch is achieved through method overriding.

Difficulty Level: H

Question 25: What is tight coupling?

Option A: Independent classes

Option B: High dependency between classes

Option C: Multiple inheritance

Option D: Abstraction

Correct Answer: High dependency between classes

Explanation: Tight coupling means classes depend heavily on each other.

Difficulty Level: M

Question 26: Which principle focuses on "what an object does" instead of "how"?

Option A: Encapsulation

Option B: Inheritance

Option C: Abstraction

Option D: Coupling

Correct Answer: Abstraction

Explanation: Abstraction focuses on functionality while hiding implementation.

Difficulty Level: M

Question 27: Which class member is shared among all objects?

Option A: Instance variable

Option B: Local variable

Option C: Static variable

Option D: Reference variable

Correct Answer: Static variable

Explanation: Static variables belong to the class rather than individual objects.

Difficulty Level: E

Question 28: Which keyword is used to implement an interface in Java?

Option A: extends

Option B: inherits

Option C: implements

Option D: interface

Correct Answer: implements

Explanation: The implements keyword allows a class to implement an interface.

Difficulty Level: E

Question 29: What happens if a subclass does not override an abstract method?

Option A: Compilation success



Option B: Runtime error

Option C: Subclass must also be abstract

Option D: JVM ignores method

Correct Answer: Subclass must also be abstract

Explanation: A subclass with unimplemented abstract methods must be declared abstract.

Difficulty Level: H

Question 30: Which OOP feature reduces redundancy?

Option A: Inheritance

Option B: Encapsulation

Option C: Coupling

Option D: Constructor

Correct Answer: Inheritance

Explanation: Inheritance avoids rewriting common code in subclasses.

Difficulty Level: E